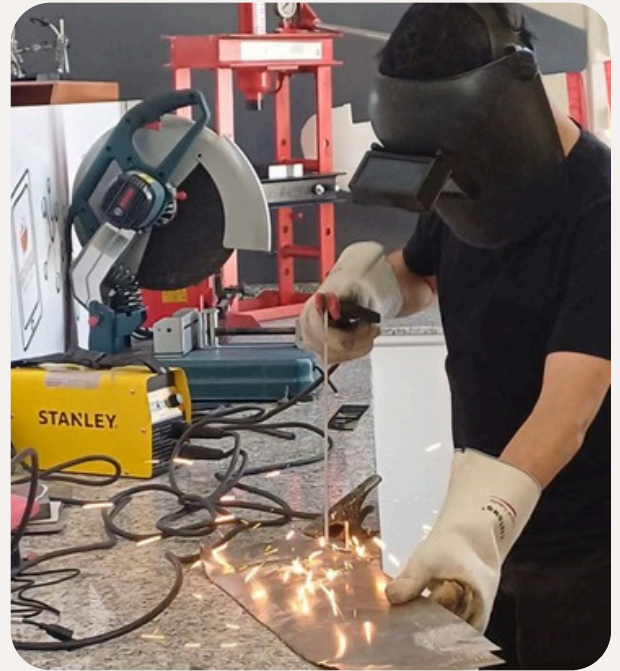


P O R T F O L I O

A B O U T M E



NAME: ABDULFETTAH TASCI

PROGRAM: MECHANICAL ENGINEERING

UNIVERSITY: ERZURUM TECHNICAL UNIVERSITY

T E C H N I C A L S U M M A R Y & M O T I V A T I O N

Mechanical Engineering student with a strong focus on UAVs, aerospace systems, and advanced engineering design. Experienced in mathematical modeling, topology optimization using nTop, fluid flow analysis with ANSYS Fluent, and structural analysis with ANSYS Mechanical. Proficient in SolidWorks, Siemens NX, ANSYS, XFLR5, and nTop for design, simulation, and performance evaluation. Also has experience working with composite materials in engineering applications. Eager to apply these skills to the development of innovative aerospace, robotics, and autonomous systems.

CONTENT

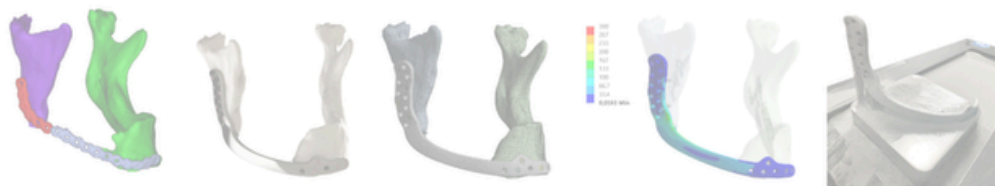


4



ADD4BIO

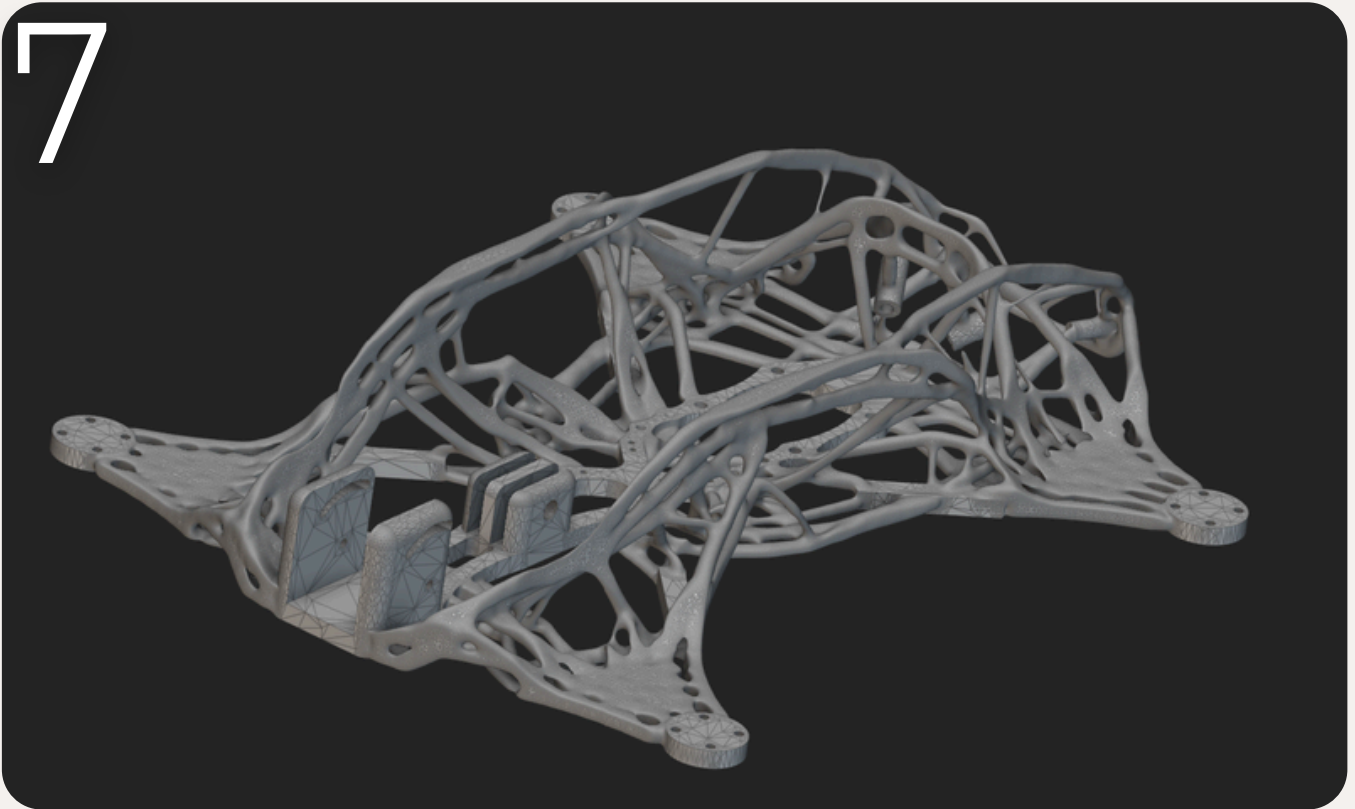
INNOVATION FOR ALL LIVES



MEDICAL ADVANCED TECHNOLOGIES CO.



CONTENT

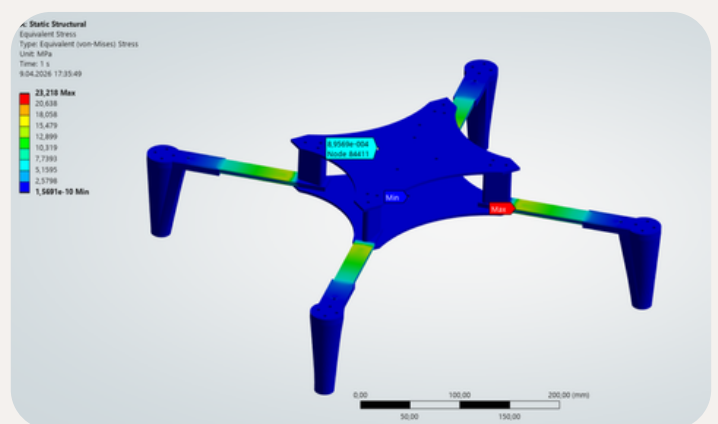
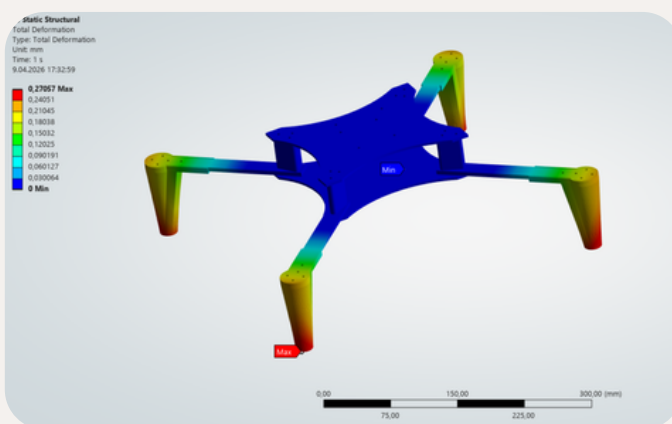
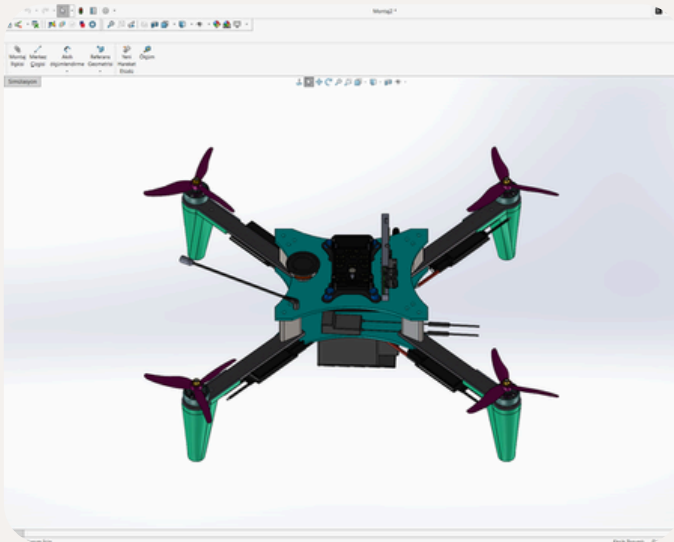


1. AUTONOMOUS SAR DRONE
2. AUTONOMOUS SAR FIXED WING
3. SURVEILLANCE AND RECONNAISSANCE UAV
4. SURGICAL ANIMATION AND TITANIUM POST-PROCESSING
| ADD4BIO
5. R&D ENGINEERING - TRAILER REAR-IMPACT TEST
SYSTEM | MAKIST
6. CFD AND STRUCTURAL ANALYSIS FOR THE TEKNOFEST
COMBAT UAV COMPETITION
7. STRUCTURAL ANALYSIS AND MANUFACTURING OF A
TOPOLOGY-INSPIRED UAV FRAME
8. CONTACT & SOCIAL

PROJECT

Autonomous Search and Rescue System (Drone)

In this project, I worked as a mechanical design engineer and was responsible for the design of the multirotor drone platform. During the design process, I considered key factors such as lightweight construction, functionality, component integration, and structural layout. This project helped me improve my experience in UAV mechanical design.



Click to watch



[Drone Flight Test \(Drive\)](#)

P R O J E C T

Autonomous Search and Rescue System (Fixed Wing)

In this project, I contributed to the design and manufacturing of the fixed-wing UAV platform for the Autonomous Search and Rescue System. This experience strengthened my skills in UAV development and mission-oriented engineering.



Click to watch

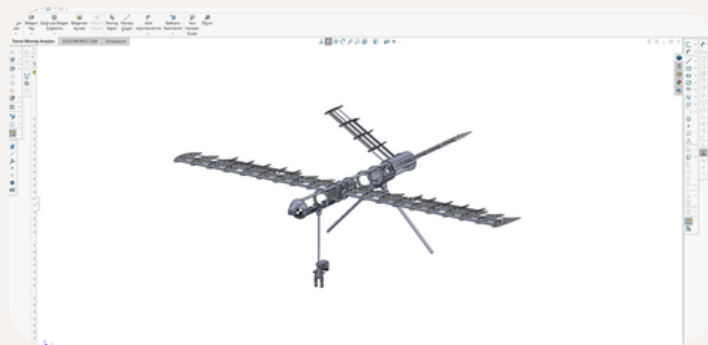


[Flight of the Fixed Wing \(on Drive\)](#)

P R O J E C T

Surveillance and Reconnaissance Drone

This section showcases a surveillance and reconnaissance drone that I designed and developed. The project focused on building a practical UAV platform for observation, monitoring, and field reconnaissance missions. It helped me strengthen my skills in UAV design, structural development, and mission-oriented engineering.



P R O J E C T

Surgical Animation and Titanium Post-Processing | Add4Bio

At Add4Bio, I created surgical animations to demonstrate the procedure pathway for the doctor and carried out the post-processing of manufactured titanium parts, including sanding and polishing operations. To maintain patient confidentiality, the animation was presented on a non-realistic skull model selected from GrabCAD rather than a real patient skull.



Click to watch

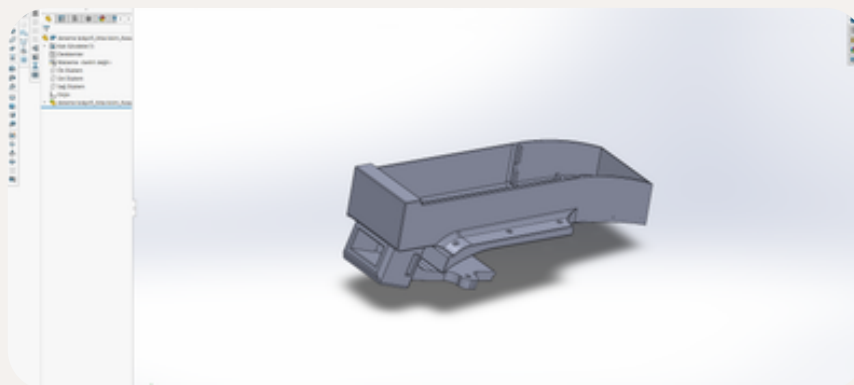
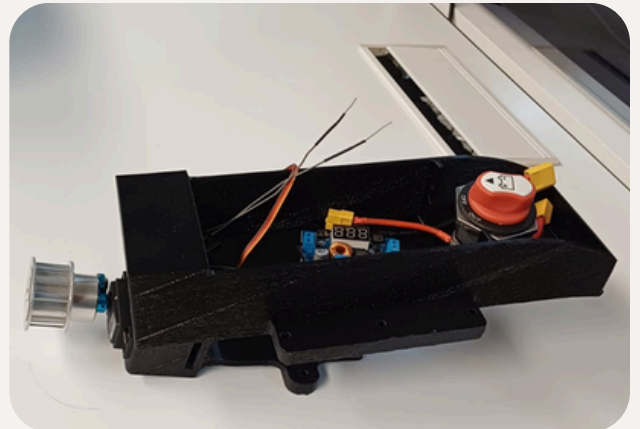


[Animation](#)

P R O J E C T

R&D Engineering – Trailer Rear-Impact Test System | Makist

I took part in a safety-focused engineering project developed to evaluate a patented energy-absorbing system mounted at the rear of truck trailers in order to reduce driver fatalities in rear-end collisions. Since the test vehicle did not contain a driver, we developed a dedicated control system for steering and a release mechanism to enable the vehicle to be guided and deployed safely during impact testing. As a member of the project team, I contributed to the development of this test setup, which supported the practical evaluation of the trailer rear-impact safety system under controlled conditions. This project strengthened my experience in mechanical system development, test engineering, and safety-oriented design.



Click to watch

► [Animation](#)

Click to watch

► [Steering Mechanism Video](#)

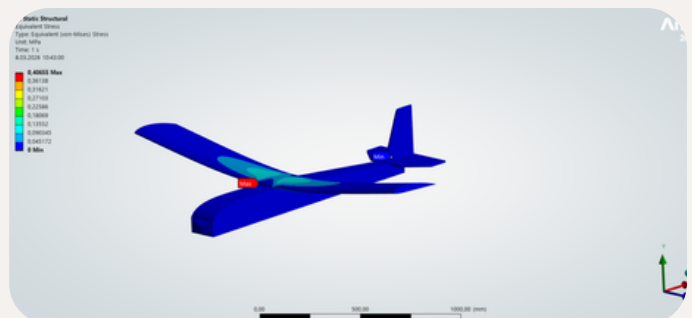
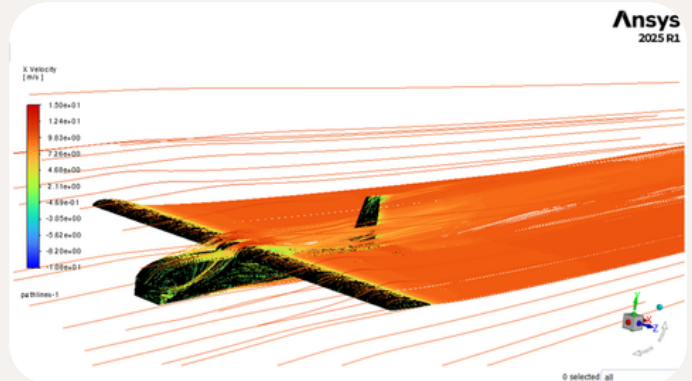
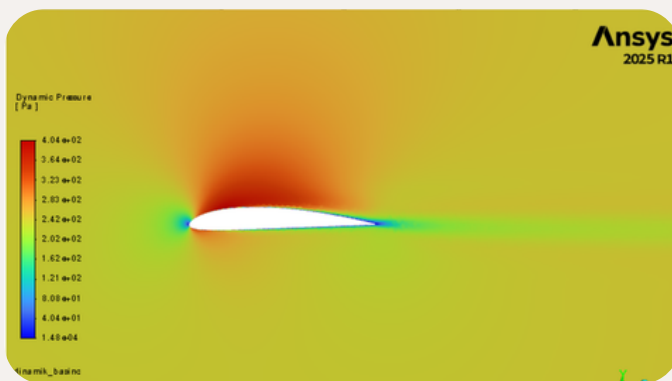
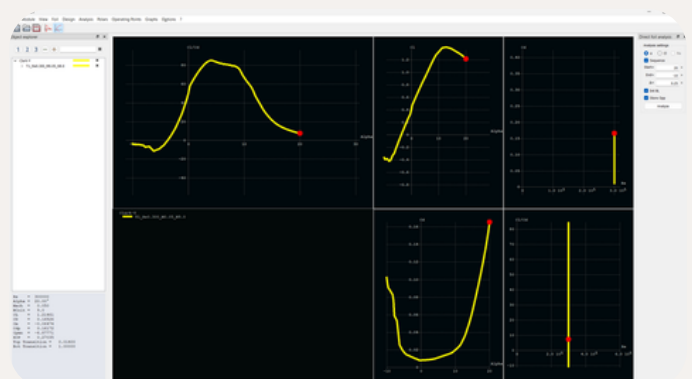
Click to watch

► [Release Mechanism](#)

PROJECT

CFD and Structural Analysis for the TEKNOFEST Combat UAV Competition

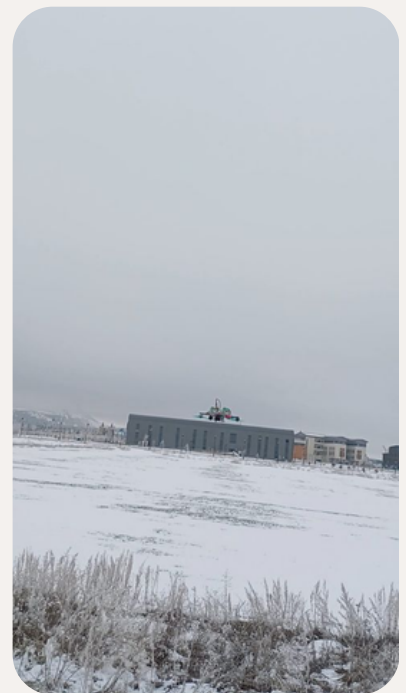
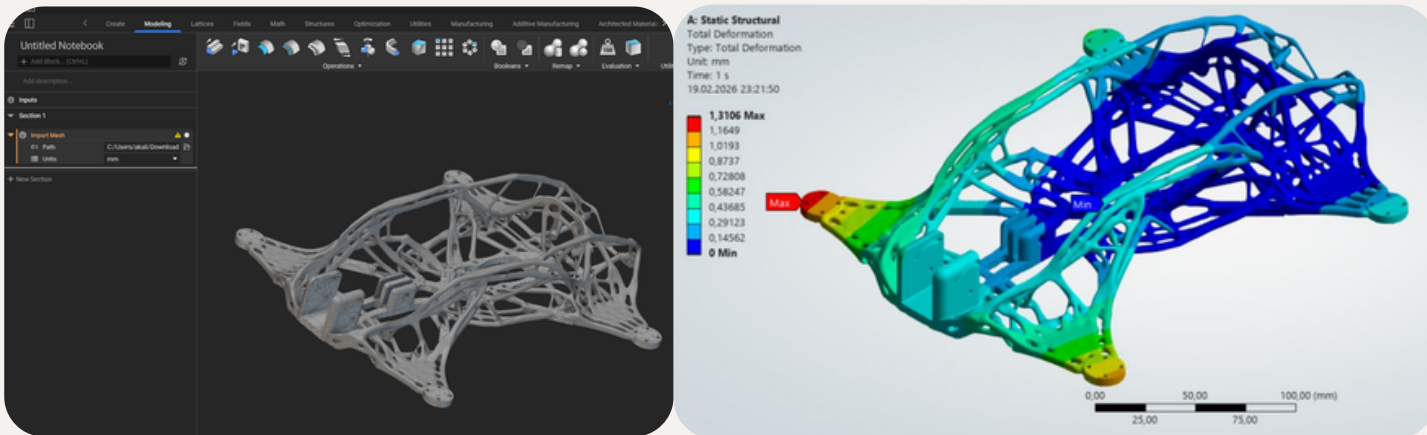
In this project, I performed the aerodynamic and structural analyses of a UAV test prototype developed for the TEKNOFEST Combat UAV Competition using ANSYS Fluent, ANSYS Mechanical, and XFLR5. The study focused on investigating the airflow behavior around the aircraft, evaluating its aerodynamic performance, and analyzing its structural response under relevant loading conditions. This project strengthened my skills in CFD, aerodynamic analysis, structural analysis, and simulation-driven UAV design.



PROJECT

Structural Analysis and Manufacturing of a Topology-Inspired UAV Frame

In this project, I developed a lightweight UAV frame by starting from a geometry model obtained from GrabCAD and evaluating its structural performance through analysis. After completing the structural analysis tests and verifying the design, we moved on to the manufacturing stage. This experience showed me how important natural and organic topologies are in lightweight engineering design, and how powerful tools like nTop can be in this process. Through this project, I became highly interested in topology-driven design and started working more deeply in this area.



[Click to watch](#)



[Flight](#)
[Video](#)

REFERENCES

Assoc. Prof. Dr. Fahri Murat – Co-Founder of Add4Bio



+90 539 827 4735



fahri.murat@erzurum.edu.tr

Prof. Dr. İrfan Kaymaz – Founder of Add4Bio | Member of the Biomechanics Research Group



irfan.kaymaz@erzurum.edu.tr

Lect. İbrahim Semih Topcuoğlu



+90 536 766 9707



ibrahim.topcuoglu@erzurum.edu.tr

CONTACT & SOCIAL



www.linkedin.com/in/abdulfettah-tasçi



<https://grabcad.com/fettah.tasçi-1>



abdulfettah.tasçi73@erzurum.edu.tr



+90 - 553 974 87 48



CUMHURİYET MAH. İNÖNÜ CAD. SEYDİKEMER/MUĞLA

ABDULFETTAH TASCI
